

Name: Yung Gadi and the Dusty Bunch

Date: 2/20/24 **Class:** 5

MyDesign Portfolio

Element D: Design Concept Generation, Analysis, and Selection

Idea 1 - Automatic door closer

Date: 2/20/24

Description: A mass hanging off of a pulley system attached to an end of the door to have it automatically close.

Drawing:



Idea 2 - Door mat

Date: 2/20/24

Description: A mat that you can wipe your feet on to collect dust from your shoes. A small enough mat can be put underneath the door, collecting dust from both sides and minimizing the dust flowing under the door

Name: Yung Gadi and the Dusty Bunch

Date: 2/20/24 **Class:** 5

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Idea 3 - Automatic Ventilation Flap Closer

Date: 2/20/24

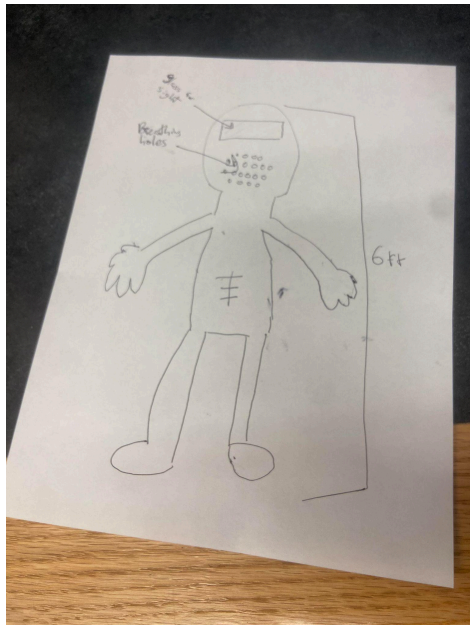
Description:

Idea 4 - hazmat suit

Date: 2/20/24

Description:

Almost like a plastic bag that sucks all the dust to prevent dust going elsewhere



using information such as scale, dimensions, and materials:

Name: Yung Gadi and the Dusty Bunch

Date: 2/20/24 **Class:** 5

Design Idea Comparison and Selection ($D1$, $D2$)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Name: Yung Gadi and the Dusty Bunch

Date: 2/20/24 **Class:** 5

Design Blueprint #1

Date:

Sketch:

Manufacturing Methods:

Step 1:

Step 2:

Step 3:

Supply List and Costs:

Design Choice	Contain dust at source	Ease of use	Maintenanc e	Dust on clothes	Room Leaka ge	Effort for setup (for us)		Tot al	Weigh ted Avera ge
Weights	25	15	15	10	15	20		100	
Automatic Door Closer	0	10	8	3	9	8		38	595
Door mat	3	9	9	2	6	10		39	655
Flap shutters (Automatic)	9	8	5	8	7	2		39	645
Cleanup Job System	6	2	2	5	5	3		23	395
Hazmat Suit	0	9	5	10	7	1		32	435

Name: Yung Gadi and the Dusty Bunch

Date: 2/20/24 **Class:** 5

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Date	2/22	2/26	2/28	3/1	3/5	3/7	3/11	3/13	3/15	3/19	3/21
Brainstorming											
Choosing a Solution											
Communicating with Mr. L											
Constructing/Obtaining solutions											
Implementing & Testing											
Finishing & Documenting											